Displacement-time graphs

AS Level | 9702 AS Ch.1 | Chapter 1: Kinematics | Lesson 1.4 | Sections 1.4

Goal: Read velocity from the gradient of a displacement–time graph, including from a tangent to a curve.

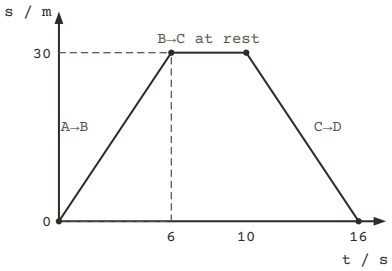
01 The big picture

The gradient of a displacement–time graph is the velocity. A straight line means constant velocity; a horizontal line means the object is at rest; a negative gradient means it is moving back towards the start.

A curved line means the velocity is changing. The instantaneous velocity is the gradient of the tangent drawn at that time.

Gradient [paraphrase]
Change in displacement ÷ change in time, read from a large triangle on the line.

Fig. 1 · A displacement-time graph



Steeper line = faster. Flat = at rest. Downward slope = moving back.

02 Key equations

Velocity = gradient

$$v = \frac{\Delta s}{\Delta t}$$

Symbols and units

s	displacement	m	t	time	s	v	velocity	$\mathrm{m\,s^{-1}}$

03 Sample problems with solutions

Sample 1 · Reading the graph
Use the graph in Fig. 1 to find the velocity during each of the three stages.

Step 1. A to B:

$$v = \frac{30 - 0}{6 - 0} = 5.0\ \mathrm{m\,s^{-1}}$$

Step 2. B to C: the line is horizontal, so the object is at rest: $v = 0$.

Step 3. C to D: the gradient is negative, so it is moving back:

$$v = \frac{0 - 30}{16 - 10} = -5.0\ \mathrm{m\,s^{-1}}$$

Answer: +5.0 m s^{-1}, then 0, then −5.0 m s^{-1} (back towards the start)

Unit check: $\frac{\mathrm{m}}{\mathrm{s}} = \mathrm{m\,s^{-1}}$ ✓

04 Practice problems

P1 What does the gradient of a displacement–time graph represent?
Answer: Velocity

P2 A straight line goes from (0 s, 0 m) to (8.0 s, 40 m). Calculate the velocity.
Answer: 5.0 m s^{-1}

P3 What does a horizontal line on a displacement–time graph show?
Answer: The object is at rest

P4 A tangent to a curved s–t graph passes through (1.0 s, 0 m) and (5.0 s, 24 m). Find the velocity there.
Answer: 6.0 m s^{-1}

05 Other useful information

Common mistakes

✗ **Wrong:** Read the velocity off the vertical axis.
✓ **Right:** The vertical axis gives displacement; velocity is the gradient.

Exam tips

- Use a large gradient triangle; a negative gradient means moving back.

What the syllabus asks

- 9702 topic 2: determine velocity using the gradient of a displacement–time graph. Coursebook 1.4.

Explore: Distance-Time Graph Explorer (simulation) · Practice questions for this lesson